

5.3.1.2 Railways

The Indian Railways is among the largest rail networks in the world. The Indian Railways route length network is spread over 1,23,236 km, with 13,452 passenger trains and 9,141 freight trains from 7,349 stations plying 23 million travellers and 3 million tonnes (MT) of freight daily. The railway network is ideal for long-distance travel and movement of bulk commodities, apart from being an energy efficient and economic mode of conveyance and transport. Indian Railways is the preferred carrier of automobiles in the country.

Government of India has focused on investing in railway infrastructure by making investor-friendly policies. It has moved quickly to enable Foreign Direct Investment (FDI) in railways to improve infrastructure for freight and high-speed trains. At present, several domestic and foreign companies are also looking to invest in Indian rail projects. Foreign Direct Investment (FDI) inflows into Railways Related Components from April 2000 to June 2020 stood at US\$ 1.12 billion.

Government initiatives

Few initiatives taken up by the Government are:

- **Freight on Priority:** Railways has embraced a “Freight on Priority” policy by pushing for an aggressive customer-centric approach to expand the freight carried not only from the traditional segments but also by attracting new customers to its fold.
- India entered a new era of mobility with Vande Bharat Express - India’s first high-tech, energy-efficient, self-propelled train. This is a prime example of the success of Make in India movement. This train will be proliferated across India and also exported globally.
- The Railways is undertaking measures for improving speed of both passenger and freight trains. Indian Railway is constructing more than 3000 km of Dedicated Freight Corridor (DFC), which would enable freight trains to run at speed of 100 kmph.
- Dedicated Freight Corridors which are special tracks and arrangements for goods trains are getting completed at unprecedented speed. They are expected to decongest the existing Indian Railway network, increase the average speed of goods trains from existing 25 to 70 kmph., run Heavy Haul trains, facilitate the running of longer (1.5 km) and double stack container trains, connect the existing ports and industrial areas for faster movement of goods, ensure an energy efficient and environment friendly rail transport system as per global standards, increase the rail share from existing 30 per cent to 45 per cent and reduce the logistic cost of transportation.
- New Services in Passenger Operations with PPP in Train Operations: Railways is now undertaking a partnership approach for passenger train operations in order to enhance overall service quality and operational efficiency. This aims at improving the passenger experience and bringing modern technologies and private investments
- India has made a huge leap of self-sufficiency in manufacturing related activities of the Railways. India is now building modern trains for itself and exporting it as well. For instance, in Uttar Pradesh only, the Locomotive

Works in Varanasi is becoming a big electric locomotive centre in India. The railway coaches which are being built here are now being exported to foreign countries also.

- A National Rail Plan (NRP) 2030 has been developed with a view to develop infrastructure by 2030 to cater to the traffic requirements upto 2050. Based on the NRP, a Vision 2024 document has been prepared to develop infrastructure by 2024 to enhance modal share of Railways in freight transportation to more than 40 per cent and to cater to the traffic requirements upto 2030.
- Electrification has been accorded high priority as a part of the national goal to transform India into a green nation. 66 per cent of track length has been electrified by November 2020. Railways aim to complete electrification of its entire broad-gauge network by 2023. The speed of electrification has been greatly scaled up from a level of 1176 km in 2014-15 to 5276 in 2018-19 and 4378 km in 2019-20.
- Government has enhanced the role of the PPP beyond providing maintenance in the areas such as redevelopment of stations, building private freight terminals and private container train operations.

5.3.1.3 Shipping and Ports

Ports are economic and service provision units of remarkable significance since they act as a place for the interchange of two transport modes, maritime and land, whether by rail or road. With a coastline of about 7,517 km. Indian ports and shipping industry plays a vital role in sustaining growth in the country's trade and commerce. According to the Ministry of Shipping, around 95 per cent of the country's trading by volume and 70 per cent by value is done through maritime transport. The country has 12 major ports and 212 notified non-major (minor/intermediate) ports along the coast-line and sea-islands. Many ports in India (such as Mundra Port, Sikka Port, Hazira Port, etc.) are evolving into specialised centres of economic activities and services and are vital to sustain future economic growth of the country. Over the last decade, there has been a steady increase in handling of cargo traffic at Indian ports (refer Table 5.3). The compound annual growth rate of total cargo throughput at Indian ports during the period 2001-02 to 2018-19 was 7.4 per cent.

Table 5.3: Growth of Cargo at Indian Ports

Parameters	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19
Trends in India's Select: Macro Parameters							
Total Cargo	2.2	4.1	8.2	1.9	5.8	6.6	6.1

Source: Based on Data from Major ports and Non-major ports.

However, Indian ports face several infrastructural and operational challenges. They include: (i) Operational efficiency of Indian ports still lags behind the global average. For instance, the turnaround time (TAT) at major ports was approximately 2.5 days in 2018-19, whereas global average benchmark is 1-2 days. (ii) last mile connectivity to the ports is one of the major constraints in smooth movement of cargo to/from the hinterland. Around 87 per cent of Indian freight uses either road or rail for transportation of goods where a significant share experiences "idle time" due to capacity constraints on highways and railway lines. Water-

borne transport which is much safer, cheaper and cleaner, compared to other modes of transportation accounts for less than 6% of India's modal split. Significant savings can be achieved by shifting movement of industrial commodities like coal, iron ore, cement and steel to coastal and inland waterways. (iii) location of industries / manufacturing centres vis-à-vis the ports is such that the country has a greater container exporting cost than that compared with competitors like China. Hence, greater savings could be possible with the presence of major manufacturing and industrial zones in coastal regions in India.

Government initiatives

The Indian Government plays an important role in supporting the ports sector. It has allowed Foreign Direct Investment (FDI) of up to 100 per cent under the automatic route for port and harbour construction and maintenance projects. Ports sector in India has received a cumulative FDI of US \$ 1.64 billion between April 2000 and March 2019. In March 2018, a revised Model Concession Agreement (MCA) was approved to make port projects more investor-friendly and make investment climate in the sector more attractive. In addition to this, project UNNATI has been started by Government of India to identify the opportunity areas for improvement in the operations of major ports. Under the project, 116 initiatives were identified out of which 91 initiatives have been implemented as of November 2018.

Sagarmala – a Port-led Development Programme

The concept of Sagarmala was approved by the Union Cabinet on 25th March 2015. As part of the programme, a National Perspective Plan (NPP) for the comprehensive development of India's 7,500 km coastline, 14,500 km of potentially navigable waterways and maritime sector has been prepared. Components of Sagarmala Programme are:

Port Modernisation and New Port Development: De-bottlenecking and capacity expansion of existing ports and development of new greenfield ports

Port Connectivity Enhancement: Enhancing the connectivity of the ports to the hinterland, optimising cost and time of cargo movement through multi-modal logistics solutions including domestic waterways (inland water transport and coastal shipping)

Port-linked Industrialisation: Developing port-proximate industrial clusters and Coastal Economic Zones to reduce logistics cost and time of EXIM and domestic cargo.

Coastal Community Development: Promoting sustainable development of coastal communities through skill development and livelihood generation activities, fisheries development, coastal tourism etc.

Coastal Shipping and Inland Waterways Transport: Impetus to move cargo through the sustainable and environment-friendly coastal and inland waterways mode.

The Indian government plans to develop 10 coastal economic regions as part of plans to revive the country's Sagarmala (string of ports) project.